A 12-bit 10GS/s Time-Interleaved SAR ADC with Even/Odd Channel-Correlated Absolute Error-Based Over-Nyquist Timing-Skew Calibration in 5nm FinFET

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2024 IEEE Symposium on VLSI Technology and Circuits (VLSI Technology and Circuits) | 979-8-3503-6146-9/24/$31.00 ©2024 IEEE | DOI: 10.1109/VLSITECHNOLOGYANDCIR46783.2024.10631314

Abstract

This paper presents an over-Nyquist 12-bit 10GS/s time-interleaved (TI) ADC with a background timing-skew calibration. An absolute error-based auto-correlation algorithm enables calibration of time skew for both the first and second Nyquist inputs, while an even and odd channel-based auto-correlation algorithm requires only three calibration steps. A prototype 10GS/s ADC in 5nm FinFET achieves a SNDR of 50.2dB and 46.2dB at input frequencies of 5GHz and 9GHz, respectively. The ADC consumes 386mW resulting in a FoM_{Walden} and $FoM_{Schreier}$ of 146.0fJ/c-s. and 151.3dB, respectively.

Keywords: SAR ADC, time interleaving, timing skew, calibration and auto-correlation

Introduction

Although time-interleaved (TI) techniques increase ADC conversion speed, they suffer from mismatch errors between interleaved channels [1-4]. The calibration of offset and gain mismatch is relatively straightforward, but timing-skew calibration presents much more challenge due to its input-dependent detection. There are several timing-skew calibration techniques based on auto-correlation algorithms that have been researched. However, these techniques have limitations, such as limiting input bandwidth up to the first Nyquist frequency and requiring increased calibration time with more channels [4,5]. This paper proposes an auto-correlation algorithm based on even and odd channel-correlated absolute error, which enables timing-skew calibration for both the first and second Nyquist inputs and requires only three calibration steps, regardless of the number of channels.

TI SAR ADC Architecture

The 12-bit 10GS/s 16-channel TI SAR ADC and timing diagram are shown in Fig. 1. It consists of a 2-stage push-pull source follower (PPSF), four ADC slices (ADC SLICE0,1,2,3), a multi-phase clock generator, a digital calibration logic, and current-voltage (IV) references. Each ADC slice is composed of one PPSF, four channels of 12-bit 625MS/s sub-SAR ADCs, and four clock re-timers with digital-controlled delay lines (DCDLs). Four sub-ADCs are placed in each ADC slice at positions k, k+4, k+8, and k+12 (where k=1,2,3,4) to prevent signal interference, such as kick-back noise, between the previous and current sub-ADC. The 10GHz main clock (ϕ_MAIN) and 625MHz multi-phase clocks (MP<1:16>) are delivered to each sub-ADC to synchronize the sampling clocks (ϕ_(S<1:16>)) with ϕ_MAIN. The DCDLs adjust the sampling clocks (ϕ_(S<1:16>)) of the sub-ADCs with a resolution of 10fs.

Absolute Error-based Timing-skew Calibration

The proposed timing-skew calibration with power detection is shown in Fig. 2. The gain and offset mismatches are digitally calibrated in digital domain, while the timing mismatches are corrected by DCDLs in analog domain. In the prototype ADC, both the conventional signed error ($e_s[i]$) and the proposed difference of absolute error ($e_a[i]$-$e_a[i-1]$) based auto-correlation algorithms are implemented. The signed error-based approach determines the direction of correction (leading or lagging) according to the polarity of the signed error to reduce timing mismatch (T_{skew}). However, the input bandwidth is limited since the polarity of the signed error is opposite between the first and second Nyquist zone. On the other hand, the proposed absolute error-based approach determines the direction of correction according to the polarity of the difference between the current and the previous absolute errors regardless of the Nyquist zone. If the difference of absolute error ($e_a[i]$-$e_a[i-1]$) is less than zero, it indicates a decrease of T_{skew}, and thus the previous direction of correction is maintained. If it is greater than zero, the direction of correction should be reversed. Meanwhile, to maintain the calibration performance even with the non-periodically present input signal such as the time division duplex (TDD), the power detection logic averages the ADC output over a defined window size (NUM_{AVG}) and compares it to the dedicated threshold level (V_{TH}). If the average of the ADC output exceeds V_{TH} (P_{OK}='1'), the results of calibration are updated; conversely, if it does not exceed (P_{OK}='0'), the previously calibrated results are frozen with the assumption of the signal absence.

Even and Odd Channel Based Auto Correlation

The proposed timing-skew calibration with even and odd channels based auto-correlation is shown in Fig. 3, with a four-channel example. The vertical solid lines indicate the input sampling clock edge of each channels (CH1~CH4) without timing mismatches, while the vertical dotted lines with the red dots indicate the real input sampling clock edge with timing mismatches, s_k (k=1,2,3,4). The timing skew of channel1 (CH1) is skew-free (s_1=0), as it is used as reference channel for calibrating timing mismatch of other channels. The proposed timing skew calibration is performed with three steps, which are the even-channel preliminary calibration, the odd-channel preliminary calibration, and the post processing steps. In the first step, the timing skew of the even channels (CH2 and CH4) is corrected based on the initial input sampling edges of the odd channels (CH1 and CH3), respectively. The input sampling edges of even channels (CH2 and CH4) are moved from the initial vertical dotted lines with red dots to the dotted lines with yellow dots, and the preliminary adjusted timing skew of even channels (a_2 and a_4) can be obtained. In the second step, the preliminary adjusted timing skew of odd channel (a_3) based on the input sampling edges of the even channels (CH2 and CH4) in the first step can be also obtained with the same manner. Finally, in the post processing step, the initial timing mismatches, s_k,

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can be calibrated simultaneously with the preliminary adjusted timing skews (a_1~a_3) of each channel, and therefore the initial vertical dotted lines of each channels with timing mismatches are aligned with the vertical solid lines with the green dots. Unlike [4], the proposed scheme only requires three calibration steps, regardless of the number of channels.

Measurement Results

The prototype ADC including an on-chip calibration engine is fabricated in a 5nm FinFET technology with an active area of 0.71mm^2, as shown in Fig. 4. Fig. 5 presents the measured dynamic performances at a sampling rate of 10GS/s, which are based on the signed and absolute error-based auto-correlation timing-skew calibration with the inputs of the first and second Nyquist zone. As shown in Fig. 6, the SNDR and SFDR are measured across sampling rates from 2GS/s to 10GS/s at a 1.6GHz input frequency with a 1.2Vpp sinusoidal input. At an operating speed of 10GS/s, the ADC achieves an effective resolution bandwidth (ERBW) of 7GHz. The power consumption of the prototype ADC is 386mW at a sampling rate of 10GS/s and the power breakdown of ADC is summarized in Fig. 6. Table I shows the performance summary and performance comparison.

References

[1] S. Kumar *et al.*, *ISSCC*, 2023, pp. 270-271. [2] A. Ali *et al.*, *ISSCC*, 2020, pp. 250-251. [3] S. Devarajan *et al.*, *ISSCC*, 2017, pp. 288-289. [4] K.-J. Moon *et al.*, *VLSI*, 2020, pp. 172-173. [5] C.-Y. Lin *et al.*, *ISSCC*, 2016, pp. 468-469.

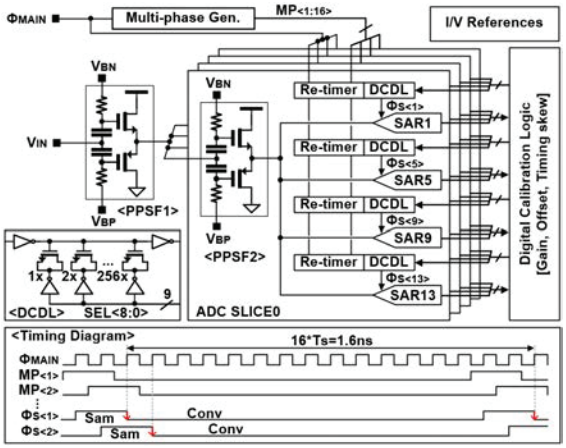


Fig. 1 Block and timing diagrams of the proposed TI-SAR ADC.

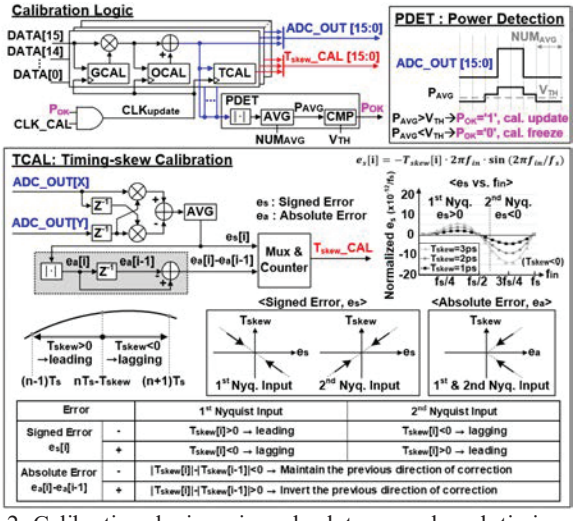


Fig. 2 Calibration logic using absolute error-based timing-skew calibration with ADC power detection.

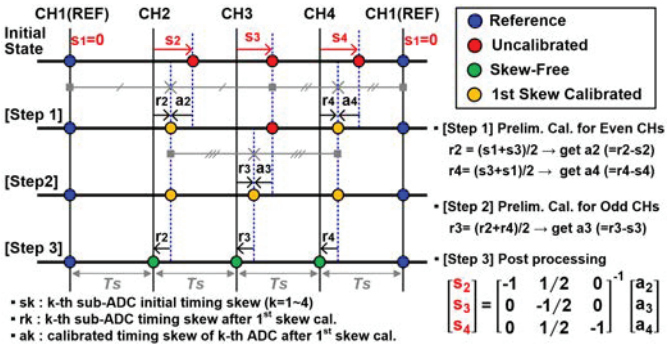


Fig. 3 Even and odd channel correlation-based timing-skew calibration (4-channel example).

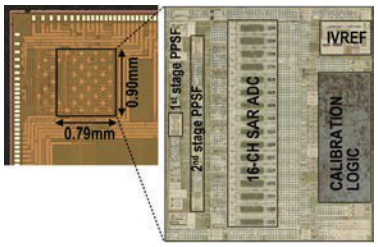


Fig. 4 Chip micrograph.

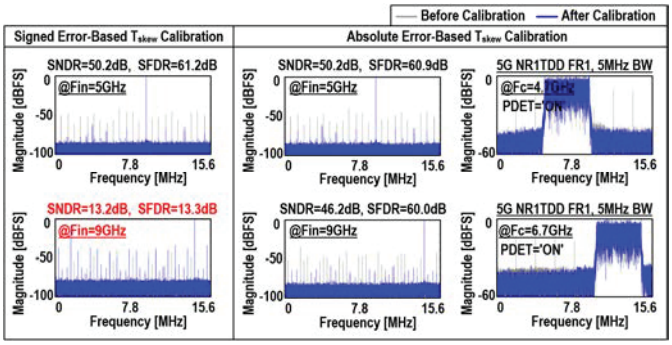
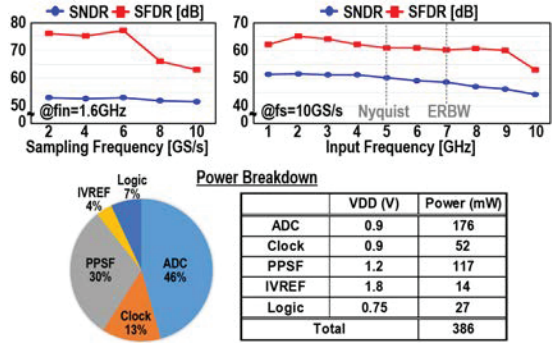


Fig. 5 Measured FFT spectrums at 10GS/s (32768-pt FFT, decimated by 321).



	VDD (V)	Power (mW)
ADC	0.9	176
Clock	0.9	52
PPSF	1.2	117
IVREF	1.8	14
Logic	0.75	27
Total		386

Fig. 6 Measured SNDR and SFDR versus sampling and input frequencies, and power breakdown.

TABLE I
PERFORMANCE SUMMARY AND COMPARISION

	This Work		ISSCC 2023 [1]	ISSCC 2020 [2]	ISSCC 2017 [3]	VLSI 2022 [4]
Technology	5nm FinFET		7nm FinFET	16nm FinFET	28nm FDSOI	5nm FinFET
Architecture	TI-SAR		TI-SAR	TI-Pipeline	TI-Pipeline	TI-Pipeline
Resolution	12		12	12	12	12
Fs [GS/s]	10		24	18	10	10
Fin [GHz]	5	9	7.24	8	4	5
SNDR at Fin [dB]	50.2	46.2	46.5	48	55	48
SFDR at Fin [dB]	61.2	60.0	76	54	64	61
Input 2nd Nyquist	O		-	O	-	-
Input TDD	O		-	-	-	-
Power [W]	0.386		0.75	1.3	2.9	0.625
Area [mm^2]	0.71		0.9	2.6	7.4	2.1
FoM_{walden} (1) [fJ/step]	146.0		181	351.9	631.2	305
$FoM_{Schreier}$ (2) [dB]	151.3		148.5	146.4	147	147

(1) FoM_{Walden}=Power/(Fs*2^{ENOB}), (2) $FoM_{Schreier}$=SNDR+10log(Fs/(2·Power))

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